ELIMINATING *PRO* FROM JAPANESE VIA NEO-DAVIDSONIAN EVENT SEMANTICS[*]

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1 Introduction

Null nominal anaphora in Japanese has been a hot topic in generative grammar. The most influential analysis of its nature arguably comes from Kuroda (1965), according to whom the Japanese lexicon includes a small *pro*, a phonologically empty but syntactically accessible pronoun. The utility of such a zero pronominal can be illustrated by the following discourse.

(1) A: *Kino* *John*$_1$***-ga*** *sara*$_2$***-o*** *wat-ta.*
yesterday John-NOM dish-ACC break-PAST
'John$_1$ broke a dish/dishes$_2$ yesterday.'
B: *Kyo-mo* ε_1 ε_2 *wat-ta-yo.*
today-too break-PAST-PRT
'ε_1 broke ε_2 today, too.'

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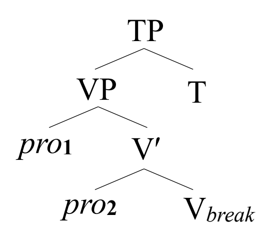

(1B) has no overt subject or object, but the empty slots ε_1 and ε_2 can be interpreted as referring to *John* and *sara* in (1A). Kuroda suggested that ε_1 and ε_2 are occupied by *pro* at the level of syntax, as in (1c). Importantly, such a null argument as he called *pro* has some crosslinguistic hallmarks of overt pronouns; it can be used, not only as a bound pronoun for quantifiers, but also as a deictic pronoun without a linguistic antecedent (e.g., Tomioka 2003, 2014, Kurafuji 2019).

This way, Kuroda's *pro* analysis has been widely accepted. In fact, however, there is reason to believe that postulating *pro* is not so innocent. Here, we will focus on its most puzzling aspect,

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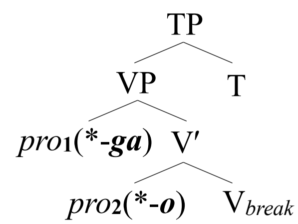
which we call *the untouchability of pro*. This is the property of *pro* that it cannot merge with any kinds of adnominal elements, including particles (e.g., cases, postpositions, focus particles) and modifiers (e.g., adjectives, genitive nouns, relative clauses). Thus, unlike overt pronouns such as *kare* 'he' and *sore* 'it', *pro* fails to host nominative *-ga* or accusative *-o*, as shown in (2).

(2)	A:	*Kino*	*kare*$_1$-***ga***	*sore*$_2$-***o***	*wat-ta.*
		yesterday	he-NOM	it-ACC	break-PAST
		'He$_1$ broke it$_2$ yesterday.'			
	B:	*Hontoni*	ε$_1$(*-***ga***)	ε$_2$(*-***o***)	*wat-ta-no*?
		really			break-PAST-Q
		'Did ε$_1$ really break ε$_2$?'			

c.



This is so mysterious a fact, since *pro* is posited as a syntactically accessible pronoun. In other words, *pro* should be able to combine with a particle in the same way as overt pronouns do, but that is impossible, and it has been unanswered why the untouchability of *pro* holds.

At this point, one might avoid this question by assuming that Japanese particles are *bound morphemes* and must attach to phonologically overt hosts. However, though that idea may readily explain the unavailability of *-ga* and *-o* in (2B), we argue that it is simply wrong, because particles can stand alone, in principle. This is particularly the case with the rule now well-known as *particle-stranding ellipsis* (PSE) (e.g., Shibata 2014, Sakamoto and Saito 2018, Sato and Maeda 2019, Takita 2020). In short, PSE allows a particle to elide its host when the host has a linguistic antecedent and the particle occurs utterance-initially, as in (3B$_1$) and (3B$_2$).

(3)	A:	[[*John*$_1$-***ga***	*sara*$_2$-***o***	*watta*]	*hi*]-*wa*	*itsu-na-no*?
		John-NOM	dish-ACC	broke	day-TOP	when-be-Q
		'When is the day that John$_1$ broke a dish/dishes$_2$?'				
	B$_1$:	[[[ε$_1$]-***Ga***	ε$_2$(*-***o***)	*watta*]	*hi*]-*wa*	*kino-desu.*
				broke	day-TOP	yesterday-be
		'The day that ε$_1$ broke ε$_2$ is yesterday.'				
	B$_2$:	[[[ε$_2$]-***O***	ε$_1$(*-***ga***)	*watta*]	*hi*]-*wa*	*kino-desu.*
				broke	day-TOP	yesterday-be
		'The day that ε$_1$ broke ε$_2$ is yesterday.'				

Thus, given that PSE is possible, Japanese particles cannot be bound morphemes; if they were, they could not even stand alone in the examples of PSE above. This means that the unavailability of *-ga* and *-o* in (2B) cannot be reduced to the notion of morphological boundness, so the untouchability of *pro* is to blame for that fact.[1]

[1] To confirm this point, it is important to note that the empty slots derived by PSE in (3) are instances of ellipsis, not of *pro*. This is clear, because as Sakamoto and Saito (2018) first pointed out, PSE are unlike overt pronouns and *pro* in requiring linguistic antecedents, and the need for them is a defining feature of ellipsis. That is, only PSE cannot be used deictically, which is illustrated by the following paradigm adapted from Sakamoto and Saito (p. 315).

(i) [Situation] Mary$_3$ is a very cute girl, and every boy in her class has a crush on her.
So, when Mary$_3$ enters the classroom in the morning, some boy says:

a.	[*Kanojo*$_3$]-***ga***	*kita*!	b.	[ε$_3$]	*kita*!	c. * [ε$_3$]-***Ga***	*kita*!
	she-NOM	came			came	-NOM	came
	'She$_3$ came!'			'ε$_3$ came!'		'ε$_3$-NOM came!'	

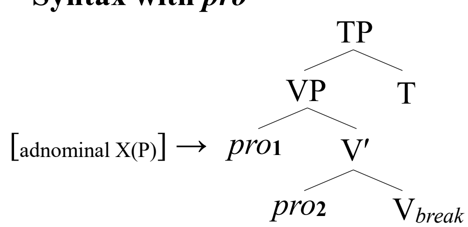
Crucially, *pro*'s untouchability can be lent stronger support from its other difference to overt pronouns noted above; that is, *pro* fails to merge with any modifiers, such as adjectives, genitive nouns, and relative clauses. For instance, consider the following discourse with relativization.

(4) A: [[*Kino* *kita*] ***kare***$_4$]-*to* [[*kyo* *kita*] ***kare***$_5$], *docchi-ga* *ii-no*?
yesterday came he-and today came he which-NOM better-Q
'Which is better, the HE$_4$ who came yesterday or the HE$_5$ who came today?'
B$_1$: *Mochiron*, [[*kyo* *kita*] ***kare***$_5$]!
of.course today came he
'Of course, the HE$_5$ who came today!'
B$_2$: * *Mochiron*, [[*kyo* *kita*] ε_5]!
of.course today came
'Of course, the ε_5 who came today!'

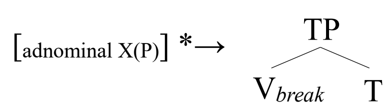
As shown in (4A), Japanese overt pronouns like *kare* can host relative clauses, but the contrast between (4B$_1$) and (4B$_2$) indicates that it is impossible for *pro*. These examples are especially telling, because they show that Japanese relative clauses involve no bound morphemes, so we cannot rely on morphological boundness to account for the unacceptability of (4B$_2$).

Now that we have good reason to believe that the untouchability of *pro* is real, what remains puzzling is how the property can be derived. We argue that the best approach to the issue is to eliminate *pro* from the syntax of Japanese. This means that the Japanese lexicon does not have any lexical items like *pro*, so *pro* simply does not exist at any level of the grammar. Importantly, our approach gives a principled account of the untouchability of *pro*, since eliminating *pro* prevents anything intended to modify it from finding a host to attach to, as shown below.

(5) a. **Syntax with *pro***



b. **Syntax without *pro***



(No nominal host to attach to)

Thus, while the *pro* analysis has difficulty ruling out the floated particles and modifiers in (2B) and (4B$_2$), our *pro*-free syntax correctly ensures that such configurations are not generable.

Of course, making a new syntax is always not so easy. The biggest challenge to us is that, when main predicates such as the verb 'break' occur with no overt case-marked nouns to become their semantic arguments, our *pro*-free syntax makes them "syntactically unsaturated," as in (5b). The question is, how can such "syntactically unsaturated" predicates be semantically interpreted? In Sections 2 and 3, we will give a full answer to it based on Williams' (2015) analysis of *implicit arguments* in English and propose a "semantic variable approach" to null nominal anaphora in Japanese, where we analyze the meaning of every Japanese predicate as filling its participant slots with *free variables*. Specifically, we will develop the idea under *Neo-Davidsonian Event Semantics* (e.g., Parsons 1990, Landman 2000), which we will show helps us to capture the same range of facts as the *pro* analysis can. Section 4 concludes this paper, suggesting that our new syntax is not so flawed, and worth developing further.

2 Empirical Basis

To set the stage for our semantic proposal, we start by arguing that Japanese null arguments have great similarities to a certain type of *implicit arguments* in English. According to Fillmore (1986), Cote (1996), Pedersen (2011), Williams (2015), Bruening (2021), and many others, a number of English transitive verbs, such as *eat* and *call*, can have implicit objects, as shown below.

(6) a. Ken's been **eating** meals all night.
b. Ken's been **eating** [ε] all night.
(Cote 1996: 166)
c. The vice-president **called** him.
d. The vice-president **called** [ε].
(Cote 1996: 123, 130)

It is also well-known that verbs of this kind are divided into at least two classes. One consists of verbs that interpret their null objects *existentially* (e.g., *eat*, *drink*, *read*, *write*, *sing*, *cook*, *bake*), and the other class includes those which interpret them *anaphorically* (e.g., *call*, *win*, *lose*, *open*, *close*, *start*, *finish*). Thus, the implicit object of *eat* is best paraphrased by an unspecified noun like *stuff*, but that of *call* is more like a pronoun or elided noun. That is, it needs to refer to a contextually given entity as in ($7B_2$), where the referent of ε_6 is identified with *John*$_6$.

(7) A: Who got John$_6$ upset?
B: Well, I guess I did. I **called** [ε_6] yesterday to cheer him$_6$ up. (Cote 1996: 140)

It then naturally follows that the use of anaphoric implicit objects is contextually restricted. For instance, they cannot be used in an out-of-the-blue context such as the following.

(8) A: What did you do today?
B: I {**ate** / #**called**} [ε]. (Cote 1996: 171)

Still, it seems true that they are much more of pronouns than elided nouns, because they can also be used deictically and refer to a salient entity in the speaker's situation of utterance.

(9) [Situation] While looking at my cellphone:
I hope John **calls** [ε] soon. (= **calls** me) (Pedersen 2011: 159)

Given these facts, it is reasonable to assume that the behavior of English implicit arguments is very close to that of Japanese null ones; both of them can be interpreted, not only anaphorically, but also deictically. It is therefore a good idea to build on empirically tenable analyses of null nominal anaphora in English to approach the nature of its Japanese counterpart.

The case study in English that we will focus on here is due to Williams (2015). The essence of his idea is to represent anaphoric implicit objects of verbs like *call*, *win*, *start*, etc. as *free variables* at the level of semantics. For instance, he suggests that the meaning of *win* expresses its implicit object as a free variable whose value is determined by an *assignment function* **g** in the utterance context. In short, such a denotation looks like the one in (10).

(10) $[\![win_7]\!]^{\mathbf{g}} = \lambda x.\ [\text{win}(x)(\mathbf{g(7)})]$

In (10) and throughout this paper, we use the notations in Heim and Krazter (1998), so for every lexical item $\mathrm{LI}_{\boldsymbol{n}}$ with a numeral index $\boldsymbol{n}$, $\mathbf{g}(\boldsymbol{n})$ stands for a free variable contained in the meaning of LI_n. Thus, (10) says that *win* has the free variable **g(7)** in its internal argument slot, but its external argument slot *x* is not saturated, thus requiring *win* to merge with one overt noun.

Note that Williams' (2015) analysis of implicit anaphors as free variables is empirically motivated. First, it correctly predicts that they can become variables bound by quantifiers. In (11a), for example, the referent of the implicit object ε_7 can vary over the contests picked out by the subject quantifier, so ε_7 can be bound the same way as the overt pronoun it_7 in (11b).

(11) a. Every contest_7 turned out to have been rigged by the person who **won** [ε_7].
b. Every contest_7 turned out to have been rigged by the person who **won** it_7.
(Williams 2015: 113)

Of course, this analogy between ε and *it* might lead one to treat ε as a pronoun that covertly occurs at the syntax. However, though Williams does not deny such an analysis, he notes that they are not fully identical. One disparity is that, when used anaphorically, pronouns require their antecedents to be *explicitly mentioned*, but implicit anaphors do not; their antecedents only need to be *easily inferable* from the context plus world knowledge. To illustrate, consider (12).

(12) a. Lee was on the schedule of presenters, and she **started** [ε] early.
b. # Lee was on the schedule of presenters, and she **started** it early.
(Williams 2015: 107)

In (12), the first clauses imply that there was a talk Lee planned to give, but that talk is not explicitly mentioned, so it can only be an antecedent of ε, making the use of *it* infelicitous. Moreover, if implicit anaphors are null pronouns, they should be part of the English lexicon and hence available to any verbs. It is not true, however, as noted by Fillmore (1986), Williams (2015), and Bruening (2021), and the contrast below between *finish* and *complete* shows that the availability of implicit anaphors is determined by the lexical idiosyncrasy of each predicate.

(13) a. There is no more need to worry about Hal's book. He has **finished** [ε].
b. * There is no more need to worry about Hal's book. He has **completed** [ε].
(Williams 2015: 111)

Thus, implicit anaphors differ from pronouns and ellipsis in three respects given in Table 1.

Types	Used deictically?	Needs explicit antecedents?	Lexically idiosyncratic?
Ellipsis	**No**	*Yes*	**No**
Pronominal	*Yes*	*Yes*	**No**
Null anaphora	*Yes*	**No**	*Yes*

Table 1: Differences between three types of anaphora in English

With Williams' (2015) analysis, we can tell these types of anaphora at least ontologically; that is, it is feasible to assume that ellipsis itself is not variables; pronouns are syntactically and semantically represented variables; and implicit anaphors are only semantically represented variables whose availability is lexically specified by a limited class of predicates.

Then, our basic hypothesis is that this lexically idiosyncratic use of free variables in English is default for Japanese, generalized over all its predicates. Specifically, we assume that every participant slot of Japanese predicates is occupied by a free variable, so their null arguments occur as only semantically represented variables. This amounts to saying that they are a lexically generalized or non-idiosyncratic version of English null anaphors, as indicated in Table 2.

Types	Used deictically?	Needs explicit antecedents?	Lexically idiosyncratic?
Ellipsis	**No**	*Yes*	**No**
Pronominal	*Yes*	*Yes*	**No**
Null anaphora	*Yes*	**No**	**No**

Table 2: Differences between three types of anaphora in Japanese

It should therefore hold that Japanese null arguments can be used with no explicit antecedents.[2] This is indeed the case, and (14) shows that the null argument ε can refer to Lee's presentation implied by the first clause, but the overt pronoun *sore* 'it' cannot.

(14) *Lee-ga kyo-no-happyosya-de, san-zi-kara* {ε /#*sore-o*} *hazimer-u-rasii-yo.*
Lee-NOM today-GEN-presenter-and 3-o'clock-from it-ACC start-PRES-heard-PRT
'I heard that Lee is the presenter today and starts {ε / it} at 3 o'clock.'

Importantly, the above treatment of Japanese null arguments also gives a solution to the semantic problem with our *pro*-free syntax, because it allows us to assume that "syntactically unsaturated" predicates such as in (5b) are in fact "semantically saturated" with free variables.

Still, this kind of approach needs further elaboration to derive the range of possible readings of what we called *pro*. For instance, Tomioka (2003, 2014) and Kurafuji (2019) note that *pro* can be used, not only as an *individual-denoting anaphor*, but also as a *predicate-denoting anaphor*. This is what we saw in (1), where ε_1 refers back to the individual 'John,' while ε_2 is understood to be a mere indefinite, such as 'a dish/dishes.' In other words, if the meaning of the Japanese verb 'break' has free variables as in (15), we need to interpret **g(2)** as denoting a predicate.

(15) $[\![\mathit{break}_{\mathbf{1,2}}]\!]^{g} = \text{break}(\mathbf{g(1)})(\mathbf{g(2)})$

Needless to say, this representation makes such an interpretation impossible, because its participant slots fulfilled with **g(1)** and **g(2)** are both for individuals (i.e., type ***e***) and cannot accommodate predicates (i.e., type **<*e, t*>**), due to type mismatch. Thus, the question for us is how we can put semantic variables of type **<*e, t*>** into the participant slots of predicates.

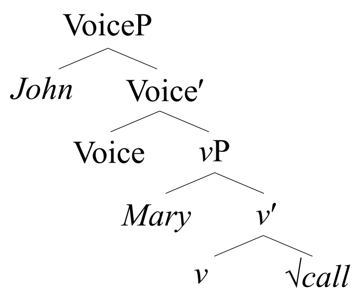
3 Theoretical Proposal

The solution that we will pursue here is to adopt *Neo-Davidsonian Event Semantics*. This is a theory of semantics established by Parsons (1990) in which the meaning of a predicate is taken to describe *events* (i.e., type ***v***) and associate them with their participants through *θ-roles*, such as *Ag(ent)*, *Th(eme)*, etc. In particular, we assume with Landman (2000) that θ-roles are functions

[2] As stated in Section 1, Japanese null arguments ε can be used deictically. See footnote 1 for an illustration.

from events to individuals (i.e., functions of type $\langle \boldsymbol{v}, \boldsymbol{e} \rangle$), so if the θ-role Ag takes an event *e* and results in the form Ag(*e*), then it denotes the individual which is the agent of *e*. To illustrate this functional treatment of θ-roles, let us consider how the predicate domain of the English sentence *John called Mary* is interpreted. Here, we adopt Basilico's (2008) analysis of verbal morphology in which a verb is syntactically decomposed into at least three (null) morphemes, namely its root √, a verbalizer *v*, and Kratzer's (1996) *Voice*, and the external and internal arguments of the root are introduced by Voice and *v*, respectively, as shown below.

(16) **English VoiceP**

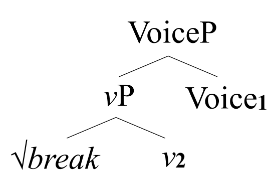


$$
\begin{aligned}
[\![\sqrt{call}]\!]^{g} &= \lambda e.\ [\text{call}(e)]\\
[\![v]\!]^{g} &= \lambda P.\lambda y.\lambda e.\ [\text{Th}(e) = y \wedge P(e)]\\
[\![v']\!]^{g} &= \lambda y.\lambda e.\ [\text{Th}(e) = y \wedge \text{call}(e)]\\
[\![Mary]\!]^{g} &= \mathbf{m}\\
[\![v\text{P}]\!]^{g} &= \lambda e.\ [\text{Th}(e) = \mathbf{m} \wedge \text{call}(e)]\\
[\![\text{Voice}]\!]^{g} &= \lambda P.\lambda x.\lambda e.\ [\text{Ag}(e) = x \wedge P(e)]\\
[\![\text{Voice}']\!]^{g} &= \lambda x.\lambda e.\ [\text{Ag}(e) = x \wedge \text{Th}(e) = \mathbf{m} \wedge \text{call}(e)]\\
[\![John]\!]^{g} &= \mathbf{j}\\
[\![\text{VoiceP}]\!]^{g} &= \lambda e.\ [\text{Ag}(e) = \mathbf{j} \wedge \text{Th}(e) = \mathbf{m} \wedge \text{call}(e)]
\end{aligned}
$$

In (16), the meaning of each phrasal node results from combining those of its daughters by Heim and Krazter's (1998) semantic rule *Function Application*, and the final output $[\![\text{VoiceP}]\!]^{g}$ says, given some event *e*, that the agent of *e* is *John*, the theme of *e* is *Mary*, and *e* is an event of calling. It is then clear that, under Landman's view, Ag(*e*) and Th(*e*) are treated as individuals (i.e., type ***e***), since the two-place relation of identity (=) identifies them with the individuals **j** and **m**. It is therefore a theoretical option to use the representations Ag(*e*) and Th(*e*) as slots that can be predicated of by variables of type $\langle \boldsymbol{e}, \boldsymbol{t} \rangle$, especially for the case of Japanese.

Given this possibility, we now propose that the Japanese morphemes of Voice and *v* for transitive verbs have the following lexical meanings which contain free variables.[3]

(17) **Japanese VoiceP**



$$
\begin{aligned}
[\![\sqrt{break}]\!]^{g} &= \lambda e.\ [\text{break}(e)]\\
[\![v_2]\!]^{g} &= \lambda P.\lambda e.\ [\mathbf{g(2)}(\text{Th}(e)) \wedge P(e)]\\
[\![v\text{P}]\!]^{g} &= \lambda e.\ [\mathbf{g(2)}(\text{Th}(e)) \wedge \text{break}(e)]\\
[\![\text{Voice}_1]\!]^{g} &= \lambda P.\lambda e.\ [\mathbf{g(1)}(\text{Ag}(e)) \wedge P(e)]\\
[\![\text{VoiceP}]\!]^{g} &= \lambda e.\ [\mathbf{g(1)}(\text{Ag}(e)) \wedge \mathbf{g(2)}(\text{Th}(e)) \wedge \text{break}(e)]
\end{aligned}
$$

As shown in (17), the Japanese Voice and *v* are lexically specified as expressions of type $\langle\langle \boldsymbol{v}, \boldsymbol{t}\rangle, \langle \boldsymbol{v}, \boldsymbol{t}\rangle\rangle$. This means that, as it stands, they introduce no individual arguments to identify with Ag(*e*) and Th(*e*), unlike the English Voice and *v* in (16), which are of type $\langle\langle \boldsymbol{v}, \boldsymbol{t}\rangle, \langle \boldsymbol{e}, \langle \boldsymbol{v}, \boldsymbol{t}\rangle\rangle\rangle$. Instead, however, the Japanese Voice and *v* contain free variables such as **g(1)** and **g(2)**, configuring them to work as functions of type $\langle \boldsymbol{e}, \boldsymbol{t} \rangle$, in the sense that they take Ag(*e*) and Th(*e*) as the target of predication. This lexical distinction of Japanese and English Voice and *v* is a reasonable one, because it can potentially account for the parametric variation between the two languages in whether the availability of null anaphors is lexically generalized or not, as discussed in Section 2.

[3] Installing semantic free variables in functional morphemes is not a new idea; see Tomioka and Kim (2017).

That is, it is feasible to assume that a limited class of v+√ combinations in English, like v+√*call*, can have the option to use free variables in describing Th(e) (cf. Bruening 2021).

Importantly, the semantics of Japanese VoiceP in (17) solves the problem noted above for the semantic variable approach. That is, it can readily derive the indefiniteness of the null theme ε_2 in (1), which gets the same meaning as the non-referential noun *a dish*/*dishes*. Suppose that the context makes the set of dishes salient and determines it as the value of **g(2)** (i.e., **g(2)** = λy. [dish(y)]). Then, the indefinite reading of the null theme ε_2 can be derived as in (18), whose final output says that Th(e) belongs to the set of dishes, so it may mean either 'a dish' or 'dishes,' given that the set consists of both single dishes and their sums (e.g., Landman 2000).

(18) $\llbracket \text{VoiceP} \rrbracket^{\mathbf{g}}$ $= \lambda e.\ [\mathbf{g(1)}(\text{Ag}(e)) \wedge \mathbf{g(2)}(\text{Th}(e)) \wedge \text{break}(e)]$
$= \lambda e.\ [\mathbf{g(1)}(\text{Ag}(e)) \wedge \boldsymbol{\lambda y.}\ \mathbf{[dish(}\boldsymbol{y}\mathbf{)]}(\text{Th}(e)) \wedge \text{break}(e)]$ ($\mathbf{g(2)} = \lambda y.\ [\text{dish}(y)]$)
$= \lambda e.\ [\mathbf{g(1)}(\text{Ag}(e)) \wedge \mathbf{dish(}\text{Th}(e)\mathbf{)} \wedge \text{break}(e)]$ (replacing y with Th(e))

On the other hand, the referent of the null agent ε_1 in (1) is understood as the individual 'John.' Getting this referential reading is totally possible if we assume that any individual of type ***e*** can be type-shifted to mean the *singleton set* that includes only that individual (e.g., Partee 1987), and that predicate-denoting variables such as **g(1)** may refer back to singleton set predicates. For instance, suppose that the value of **g(1)** is determined as the singleton set of the contextually salient individual 'John' (i.e., **g(1)** = λx. [x = **j**]). Then, the referential reading of the null agent ε_1 can be derived as in (19), whose final output says that Ag(e) is 'John.'

(19) $\llbracket \text{VoiceP} \rrbracket^{\mathbf{g}}$ $= \lambda e.\ [\mathbf{g(1)}(\text{Ag}(e)) \wedge \mathbf{dish(}\text{Th}(e)\mathbf{)} \wedge \text{break}(e)]$
$= \lambda e.\ [\boldsymbol{\lambda x.}\ \mathbf{[}\boldsymbol{x}\mathbf{ = j]}(\text{Ag}(e)) \wedge \mathbf{dish(}\text{Th}(e)\mathbf{)} \wedge \text{break}(e)]$ ($\mathbf{g(1)} = \lambda x.\ [x = \mathbf{j}]$)
$= \lambda e.\ [\text{Ag}(e) \mathbf{= j} \wedge \mathbf{dish(}\text{Th}(e)\mathbf{)} \wedge \text{break}(e)]$ (replacing x with Ag(e))

Thus, the "syntactically unsaturated" VoiceP in (17) has no argument slots to semantically saturate, and its interpretation is successfully done with the aid of its equipped free variables.

At this point, one might wonder how such a VoiceP introduces overt arguments such as *John* and *sara* in (1). For that question, we assume that the free variables can be bound by λ-operators via a version of Heim and Krazter's (1998) semantic rule *Predicate Abstraction* (PA). Specifically, we propose to revise the nature of PA as a type-shifting rule in the following way.

(20) If a node α dominates a node $\beta_{\boldsymbol{n}}$ indexed with a numeral ***n***,
then for any assignment **g**, $\llbracket {}^{\mathbf{PA}}\alpha \rrbracket^{\mathbf{g}} = \boldsymbol{\lambda X.}\ \llbracket \alpha \rrbracket^{\mathbf{g}[\boldsymbol{n}\rightarrow \boldsymbol{X}]}$,
where $\mathbf{g}[\boldsymbol{n}\rightarrow\boldsymbol{X}]$ is the **g** modified so as to assign ***X*** to ***n***.

This version of PA may apply to VoiceP and vP, since they dominate a node indexed with a numeral, namely Voice_1 and v_2. Thus, if applied to VoiceP, our PA indicates the result as $^{\mathbf{PA}}$VoiceP and turns the semantic type of VoiceP from **<*v*, *t*>** into **<<*e*, *t*>, <*v*, *t*>>**, as shown below.

(21) $^{\mathbf{PA}}$VoiceP
vP Voice_1

$\llbracket {}^{\mathbf{PA}}\text{VoiceP} \rrbracket^{\mathbf{g}}$ $= \boldsymbol{\lambda X.}\ \llbracket \text{VoiceP} \rrbracket^{\mathbf{g}[\mathbf{1}\rightarrow \boldsymbol{X}]}$
$= \boldsymbol{\lambda X.}\lambda e.\ [\mathbf{g}^{[\mathbf{1}\rightarrow \boldsymbol{X}]}\mathbf{(1)}(\text{Ag}(e)) \wedge \mathbf{g}^{[\mathbf{1}\rightarrow \boldsymbol{X}]}\mathbf{(2)}(\text{Th}(e)) \wedge \text{break}(e)]$
$= \boldsymbol{\lambda X.}\lambda e.\ [\boldsymbol{X}(\text{Ag}(e)) \wedge \mathbf{g}^{[\mathbf{1}\rightarrow \boldsymbol{X}]}\mathbf{(2)}(\text{Th}(e)) \wedge \text{break}(e)]$

With this VoiceP, it is now possible to semantically combine the subject *John*, because it can be interpreted as a predicate of type $\boldsymbol{<e, t>}$ (i.e., $[\![John]\!]^{g} = \lambda x.\ [x = \mathbf{j}]$). Their composition is described in (22), and the same introduction process may be extended to the object *sara*.[4]

(22) VoiceP′
John ${}^{\mathbf{PA}}$VoiceP

$$[\![\text{VoiceP}']\!]^{g} = \boldsymbol{\lambda X}.\lambda e.\ [\boldsymbol{X}(\text{Ag}(e)) \wedge \mathbf{g}^{[\mathbf{1}\to \boldsymbol{X}]}(\mathbf{2})(\text{Th}(e)) \wedge \text{break}(e)]([\![John]\!]^{g})$$
$$= \lambda e.\ [[\![John]\!]^{g}(\text{Ag}(e)) \wedge \mathbf{g}^{[\mathbf{1}\to \boldsymbol{X}]}(\mathbf{2})(\text{Th}(e)) \wedge \text{break}(e)]$$
$$= \lambda e.\ [\text{Ag}(e) = \mathbf{j} \wedge \mathbf{g}^{[\mathbf{1}\to \boldsymbol{X}]}(\mathbf{2})(\text{Th}(e)) \wedge \text{break}(e)]$$

This way, our semantic variable approach to *pro* can be maintained if we implement it in tandem with such frameworks as Neo-Davidsonian Event Semantics, which provides a successful means of associating the participant slots of predicates with free variables of type $\boldsymbol{<e, t>}$.

4 Conclusion

In this paper, we have proposed a *pro*-free syntax and a free-variable semantics for null nominal anaphora in Japanese. Importantly, our proposal has a nontrivial theoretical consequence. It is to confirm Kuroda's (1988) thesis that Japanese lacks the *Extended Projection Principle* (EPP), which requires every finite TP to have a noun at its edge. This is a natural outcome, because our theory allows verbs to introduce no such noun as TP can attract, as in (17). Though Miyagawa's (2010) seminal work has argued for the EPP in Japanese based on the scope relation between negation and the subject quantifier, evidence to the contrary has also been accumulated, as Shibata (2015) notices. Here, we cannot add new support to our position, but refer the reader to Tanaka's (2016) argument, which examines the distribution of the genuine NPI *sonnani* 'so.'

Finally, we note that eliminating *pro* from Japanese potentially opens up a new way to look at the nature of scrambling in Japanese. In fact, we have pursued such a possibility in Shimamura and Tanaka (to appear) and attempted to dispense with scrambling as an independent movement operation. Still, the proposal we made in that paper differs in some technical details from that we put forth in the present study. A remaining task for us is to consider how they can be unified.

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[4] Our semantic system for Japanese also allows quantifiers to combine with VoiceP or *v*P if we revise the semantic type of quantifiers and consider it, not as $\boldsymbol{<<e, <v, t>>, <v, t>>}$, but as $\boldsymbol{<<<e, t>, <v, t>>, <v, t>>}$. For example, we assume that the universal quantifier *zen'in* 'everyone' has the following denotation.

(i) $[\![zen'in]\!]^{g} = \lambda R_{<<e,t>,<v,t>>}.\lambda e.\forall X.\ [X \subseteq [\![\text{person}]\!]^{g} \wedge |X| = 1 \to \exists e'.\ [e' \leq e \wedge R(X)(e')]]$

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